National University of Computer and Emerging Sciences, Islamabad
Department of Computer Science



Inventory & Stock Management System

Project Team

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Fully Dressed Use cases

Use Case 1: User Login

<table>
<tr><td>Use Case Section</td><td>Comment</td></tr>
<tr><td>Use Case Name</td><td>User Login</td></tr>
<tr><td>Scope</td><td>Inventory & Stock Management System</td></tr>
<tr><td>Level</td><td>User Goal</td></tr>
<tr><td>Primary Actor</td><td>Admin/Staff</td></tr>
<tr><td>Stakeholders and Interests</td><td>Admin/Staff: wants secure system access.
Admin: ensures authorized users only.</td></tr>
<tr><td>Main Success Scenario</td><td>
<table>
<tr><th>Actor Action</th><th>System Response</th></tr>
<tr><td>User launches application.</td><td>System displays login screen.</td></tr>
<tr><td>User enters username/password and clicks Login.</td><td>System validates credentials.</td></tr>
<tr><td>Credentials valid.</td><td>System grants access to appropriate dashboard.</td></tr>
</table>
</td></tr>
<tr><td>Extensions</td><td>Invalid credentials → system shows 'Invalid username or password'.</td></tr>
<tr><td>Preconditions</td><td>User must be registered with valid credentials.</td></tr>
<tr><td>Postconditions</td><td>User successfully logged in.</td></tr>
</table>

Use Case 2: Add New Product

<table>
<tr><td>Use Case Section</td><td>Comment</td></tr>
<tr><td>Use Case Name</td><td>Add New Product</td></tr>
<tr><td>Scope</td><td>Inventory & Stock Management System</td></tr>
<tr><td>Level</td><td>User Goal</td></tr>
<tr><td>Primary Actor</td><td>Admin</td></tr>
<tr><td>Stakeholders and Interests</td><td>Admin: maintains product catalog.
Staff: uses products for stock tracking.</td></tr>
<tr><td>Main Success Scenario</td><td>
<table>
<tr><th>Actor Action</th><th>System Response</th></tr>
<tr><td>Admin opens Product Management > Add Product.</td><td>System shows product entry form.</td></tr>
<tr><td>Admin enters product details and submits.</td><td>System validates and saves product.</td></tr>
<tr><td>Valid details provided.</td><td>System confirms successful addition.</td></tr>
</table>
</td></tr>
<tr><td>Extensions</td><td>Missing fields → highlight and prompt correction.</td></tr>
<tr><td>Preconditions</td><td>Admin logged in; product not already exists.</td></tr>
<tr><td>Postconditions</td><td>New product added to database.</td></tr>
</table>

Use Case 3: Update Product Details

<table>
<tr><td>Use Case Section</td><td>Comment</td></tr>
<tr><td>Use Case Name</td><td>Update Product Details</td></tr>
<tr><td>Scope</td><td>Inventory & Stock Management System</td></tr>
<tr><td>Level</td><td>User Goal</td></tr>
<tr><td>Primary Actor</td><td>Admin</td></tr>
<tr><td>Stakeholders and Interests</td><td>Admin: maintains accurate and updated product info.</td></tr>
<tr><td>Main Success Scenario</td><td>
<table>
<tr><th>Actor Action</th><th>System Response</th></tr>
<tr><td>Admin searches for product.</td><td>System shows product details.</td></tr>
<tr><td>Admin edits and saves changes.</td><td>System validates and updates database.</td></tr>
<tr><td>Changes valid.</td><td>System confirms update.</td></tr>
</table>
</td></tr>
<tr><td>Extensions</td><td>Invalid data → system rejects changes.</td></tr>
<tr><td>Preconditions</td><td>Admin logged in and product exists.</td></tr>
<tr><td>Postconditions</td><td>Product details updated.</td></tr>
</table>

Use Case 4: Record Stock Inflow

<table>
<tr><td>Use Case Section</td><td>Comment</td></tr>
<tr><td>Use Case Name</td><td>Record Stock Inflow</td></tr>
<tr><td>Scope</td><td>Inventory & Stock Management System</td></tr>
<tr><td>Level</td><td>User Goal</td></tr>
<tr><td>Primary Actor</td><td>Staff</td></tr>
<tr><td>Stakeholders and Interests</td><td>Staff: records received goods.
Admin: keeps accurate total stock.</td></tr>
<tr><td>Main Success Scenario</td><td>
<table>
<tr><th>Actor Action</th><th>System Response</th></tr>
<tr><td>Staff selects 'Stock Inflow' option.</td><td>System shows inflow form.</td></tr>
<tr><td>Staff enters product and quantity.</td><td>System validates entry and updates stock.</td></tr>
<tr><td>Entry valid.</td><td>System confirms addition and logs transaction.</td></tr>
</table>
</td></tr>
<tr><td>Extensions</td><td>Quantity invalid → show error.</td></tr>
<tr><td>Preconditions</td><td>Staff logged in, product exists.</td></tr>
<tr><td>Postconditions</td><td>Stock quantity increased, transaction logged.</td></tr>
</table>

Use Case 5: Process Stock Outflow

<table>
<tr><td>Use Case Section</td><td>Comment</td></tr>
<tr><td>Use Case Name</td><td>Process Stock Outflow</td></tr>
<tr><td>Scope</td><td>Inventory & Stock Management System</td></tr>
<tr><td>Level</td><td>User Goal</td></tr>
<tr><td>Primary Actor</td><td>Staff</td></tr>
<tr><td>Stakeholders and Interests</td><td>Staff: records sales or damaged goods.
Admin: monitors stock deductions.</td></tr>
<tr><td>Main Success Scenario</td><td>
<table>
<tr><th>Actor Action</th><th>System Response</th></tr>
<tr><td>Staff selects 'Stock Outflow'.</td><td>System shows outflow form.</td></tr>
<tr><td>Staff enters product, quantity, and reason.</td><td>System validates and deducts stock.</td></tr>
<tr><td>Sufficient stock available.</td><td>System confirms deduction and logs event.</td></tr>
</table>
</td></tr>
<tr><td>Extensions</td><td>Insufficient stock → system shows 'Not enough quantity'.</td></tr>
<tr><td>Preconditions</td><td>Staff logged in, sufficient stock available.</td></tr>
<tr><td>Postconditions</td><td>Stock decreased, transaction recorded.</td></tr>
</table>

Use Case 6: Check Low Stock Alerts

<table>
<tr><td>Use Case Section</td><td>Comment</td></tr>
<tr><td>Use Case Name</td><td>Check Low Stock Alerts</td></tr>
<tr><td>Scope</td><td>Inventory & Stock Management System</td></tr>
<tr><td>Level</td><td>User Goal</td></tr>
<tr><td>Primary Actor</td><td>Admin/Staff</td></tr>
<tr><td>Stakeholders and Interests</td><td>Staff/Admin: monitors low-stock items to reorder timely.</td></tr>
<tr><td>Main Success Scenario</td><td>
<table>
<tr><th>Actor Action</th><th>System Response</th></tr>
<tr><td>User opens dashboard.</td><td>System checks all stock levels.</td></tr>
<tr><td>Threshold crossed for any item.</td><td>System lists items in 'Low Stock Alerts'.</td></tr>
</table>
</td></tr>
<tr><td>Extensions</td><td>No low-stock items → show 'All stock levels sufficient'.</td></tr>
<tr><td>Preconditions</td><td>User logged in, products have thresholds.</td></tr>
<tr><td>Postconditions</td><td>Low-stock items displayed on dashboard.</td></tr>
</table>

Use Case 7: Add New Supplier

<table>
<tr><td>Use Case Section</td><td>Comment</td></tr>
<tr><td>Use Case Name</td><td>Add New Supplier</td></tr>
<tr><td>Scope</td><td>Inventory & Stock Management System</td></tr>
<tr><td>Level</td><td>User Goal</td></tr>
<tr><td>Primary Actor</td><td>Admin</td></tr>
<tr><td>Stakeholders and Interests</td><td>Admin: maintains supplier database for product linking.</td></tr>
<tr><td>Main Success Scenario</td><td>
<table>
<tr><th>Actor Action</th><th>System Response</th></tr>
<tr><td>Admin navigates to Supplier Management.</td><td>System shows add supplier form.</td></tr>
<tr><td>Admin fills supplier details and submits.</td><td>System validates and saves record.</td></tr>
<tr><td>All valid.</td><td>System confirms supplier added.</td></tr>
</table>
</td></tr>
<tr><td>Extensions</td><td>Missing data → prompt correction.</td></tr>
<tr><td>Preconditions</td><td>Admin logged in; supplier not duplicate.</td></tr>
<tr><td>Postconditions</td><td>New supplier added to system.</td></tr>
</table>

Use Case 8: Generate Stock Summary Report

<table>
<tr><td>Use Case Section</td><td>Comment</td></tr>
<tr><td>Use Case Name</td><td>Generate Stock Summary Report</td></tr>
<tr><td>Scope</td><td>Inventory & Stock Management System</td></tr>
<tr><td>Level</td><td>User Goal</td></tr>
<tr><td>Primary Actor</td><td>Admin</td></tr>
<tr><td>Stakeholders and Interests</td><td>Admin: wants overview of stock for decisions.</td></tr>
<tr><td>Main Success Scenario</td><td>
<table>
<tr><th>Actor Action</th><th>System Response</th></tr>
<tr><td>Admin selects 'Stock Summary Report'.</td><td>System retrieves stock data.</td></tr>
<tr><td>Admin requests display.</td><td>System generates and displays summary.</td></tr>
</table>
</td></tr>
<tr><td>Extensions</td><td>No data → show 'No records available'.</td></tr>
<tr><td>Preconditions</td><td>Admin logged in.</td></tr>
<tr><td>Postconditions</td><td>Report displayed on screen.</td></tr>
</table>

Use Case 9: Generate Purchase History Report

<table>
<tr><td>Use Case Section</td><td>Comment</td></tr>
<tr><td>Use Case Name</td><td>Generate Purchase History Report</td></tr>
<tr><td>Scope</td><td>Inventory & Stock Management System</td></tr>
<tr><td>Level</td><td>User Goal</td></tr>
<tr><td>Primary Actor</td><td>Admin</td></tr>
<tr><td>Stakeholders and Interests</td><td>Admin: analyzes inflow over time for trends.</td></tr>
<tr><td>Main Success Scenario</td><td>
<table>
<tr><th>Actor Action</th><th>System Response</th></tr>
<tr><td>Admin selects 'Purchase History Report'.</td><td>System prompts date range.</td></tr>
<tr><td>Admin enters range.</td><td>System compiles inflow transactions and displays report.</td></tr>
</table>
</td></tr>
<tr><td>Extensions</td><td>Invalid date range → system prompts correction.</td></tr>
<tr><td>Preconditions</td><td>Admin logged in.</td></tr>
<tr><td>Postconditions</td><td>Purchase report generated and displayed.</td></tr>
</table>

Use Case 10: Search for a Product

<table>
<tr><td>Use Case Section</td><td>Comment</td></tr>
<tr><td>Use Case Name</td><td>Search for a Product</td></tr>
<tr><td>Scope</td><td>Inventory & Stock Management System</td></tr>
<tr><td>Level</td><td>User Goal</td></tr>
<tr><td>Primary Actor</td><td>Admin/Staff</td></tr>
<tr><td>Stakeholders and Interests</td><td>Admin/Staff: quickly find product details.</td></tr>
<tr><td>Main Success Scenario</td><td><table>
<tr><th>Actor Action</th><th>System Response</th></tr>
<tr><td>User enters search keyword.</td><td>System queries database.</td></tr>
<tr><td>Matches found.</td><td>System displays results list.</td></tr>
</table></td></tr>
<tr><td>Extensions</td><td>No matches → system shows 'No products found'.</td></tr>
<tr><td>Preconditions</td><td>User logged in.</td></tr>
<tr><td>Postconditions</td><td>Matching products displayed.</td></tr>
</table>

Use Case 11: Manage User Accounts

<table>
<tr><td>Use Case Section</td><td>Comment</td></tr>
<tr><td>Use Case Name</td><td>Manage User Accounts</td></tr>
<tr><td>Scope</td><td>Inventory & Stock Management System</td></tr>
<tr><td>Level</td><td>Summary</td></tr>
<tr><td>Primary Actor</td><td>Admin</td></tr>
<tr><td>Stakeholders and Interests</td><td>Admin: manages system access and permissions.</td></tr>
<tr><td>Main Success Scenario</td><td>
<table>
<tr><th>Actor Action</th><th>System Response</th></tr>
<tr><td>Admin opens User Management.</td><td>System lists all users.</td></tr>
<tr><td>Admin creates/updates/deactivates user.</td><td>System validates and saves changes.</td></tr>
</table>
</td></tr>
<tr><td>Extensions</td><td>Admin attempts to deactivate self → show 'Action not allowed'.</td></tr>
<tr><td>Preconditions</td><td>Admin logged in.</td></tr>
<tr><td>Postconditions</td><td>User accounts updated.</td></tr>
</table>

Use Case 12: Manage Product Categories

<table>
<tr><td>Use Case Section</td><td>Comment</td></tr>
<tr><td>Use Case Name</td><td>Manage Product Categories</td></tr>
<tr><td>Scope</td><td>Inventory & Stock Management System</td></tr>
<tr><td>Level</td><td>Summary</td></tr>
<tr><td>Primary Actor</td><td>Admin</td></tr>
<tr><td>Stakeholders and Interests</td><td>Admin: organizes and maintains product categories.</td></tr>
<tr><td>Main Success Scenario</td><td>
<table>
<tr><th>Actor Action</th><th>System Response</th></tr>
<tr><td>Admin opens Category Management.</td><td>System lists categories.</td></tr>
<tr><td>Admin adds, edits, or deletes category.</td><td>System validates and updates database.</td></tr>
</table>
</td></tr>
<tr><td>Extensions</td><td>Category in use → prompt reassignment before deletion.</td></tr>
<tr><td>Preconditions</td><td>Admin logged in.</td></tr>
<tr><td>Postconditions</td><td>Categories added/edited/deleted.</td></tr>
</table>